

Debt Expansion Restraint and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Debt Expansion Restraint (DER), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on DER achieves an annualized gross (net) Sharpe ratio of 0.48 (0.37), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 25 (19) bps/month with a t-statistic of 3.22 (2.54), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets) is 22 bps/month with a t-statistic of 3.01.

1 Introduction

Asset prices reflect investors' marginal utility across states and time, with cross-sectional return differences arising from systematic exposure to consumption risk. While the consumption-based asset pricing framework provides a unified theory for understanding risk premia, empirical evidence linking firm characteristics to consumption risk remains incomplete. We identify a critical gap in understanding how firms' debt expansion decisions relate to their exposure to aggregate consumption shocks and marginal utility fluctuations. Debt Expansion Restraint (DER), measured as the negative change in long-term debt scaled by total assets, captures firms' conservative financing behavior that potentially signals lower exposure to consumption disasters and macroeconomic downturns.

Firms exhibiting debt expansion restraint maintain financial flexibility that reduces their sensitivity to aggregate consumption shocks, particularly during economic contractions when marginal utility is high (Campbell and Cochrane, 1999; Bansal and Yaron, 2004). The consumption-based asset pricing framework suggests that assets providing insurance against consumption declines command lower expected returns, as investors accept reduced compensation for holdings that hedge adverse consumption states (Breedon, 1979; Cochrane, 2005). Conservative debt policies enable firms to better navigate periods of high marginal utility when external financing becomes costly and consumption growth deteriorates, effectively reducing their covariance with the stochastic discount factor (Lettau and Ludvigson, 2001). During economic disasters characterized by sharp consumption contractions, financially flexible firms experience smaller valuation losses and maintain operational continuity, while highly leveraged competitors face distress costs that amplify their exposure to systematic consumption risk (Barro, 2006; Wachter, 2013). The state-dependent nature of debt expansion's impact manifests through asymmetric exposure to long-run consumption risks, where conservative firms' cash flows exhibit lower sensitivity to

persistent consumption shocks that drive the equity premium (Bansal and Yaron, 2004; Hansen et al., 2008).

The DER strategy generates an annualized gross Sharpe ratio of 0.48 and achieves a monthly average abnormal gross return of 25 basis points relative to the Fama-French six-factor model with a t-statistic of 3.22. Among the largest stocks exceeding the 80th NYSE percentile, the DER strategy delivers an average return of 34 basis points per month with a t-statistic of 3.65, demonstrating robust performance that extends beyond small-cap securities. The strategy’s gross monthly alpha of 22 basis points remains highly significant with a t-statistic of 3.01 when controlling for the six Fama-French factors plus six closely related anomalies including net debt financing and change in net operating assets. Net of transaction costs, the DER strategy achieves an annualized Sharpe ratio of 0.37, ranking in the 94th percentile among 212 documented anomalies in the factor zoo. The long-short portfolio loads significantly on the CMA factor with a coefficient of 0.24 and t-statistic of 4.66, consistent with debt-constrained firms’ reduced investment sensitivity to consumption growth fluctuations. Conditional double sorts reveal that DER’s predictive power persists across all size quintiles, with the strategy generating significant alphas ranging from 25 to 36 basis points per month among large-cap stocks.

Our study extends the consumption-based asset pricing literature by documenting how debt expansion restraint captures cross-sectional variation in firms’ exposure to consumption risk, complementing recent work on financial constraints and expected returns (Whited and Wu, 2006; Livdan et al., 2009). Unlike Cooper et al. (2008) who examine asset growth’s relation to returns through investment frictions, we demonstrate that debt financing decisions specifically reflect firms’ hedging capacity against consumption disasters and state-dependent marginal utility shocks. While Lyandres et al. (2007) document the return predictability of external financing activities, our focus on debt expansion restraint isolates the consumption risk

channel from mechanical dilution effects and reveals stronger performance among large-cap stocks where consumption-based pricing should dominate. We advance the methodological frontier by showing that DER maintains significant predictive power after controlling for prominent investment and financing anomalies, suggesting that conservative debt policies capture a distinct dimension of consumption risk exposure not subsumed by existing factors (Hou et al., 2015). The robust performance of DER across market capitalizations and its resilience to transaction costs provide practical implications for implementing consumption-based investment strategies. Our findings illuminate the mechanism through which corporate financing decisions embed information about systematic consumption risk, offering new evidence supporting the fundamental predictions of consumption-based asset pricing theory.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our analysis focuses on annual observations from the COMPUSTAT Annual Industrial and Research files, covering both active and inactive firms to mitigate survivorship bias. The sample spans several decades of financial data, allowing us to examine debt issuance patterns across various market conditions and business cycles. construction of our Debt Expansion Restraint signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the dollar amount of long-term debt issued during the fiscal year, capturing proceeds from the issuance of bonds, notes, and other long-term borrowing instruments. This flow variable reflects a firm’s external financing activities through debt markets during the reporting period. Total assets (AT) measures the book value of all assets reported on the balance sheet at fiscal year-end, providing a comprehensive measure of firm

size that includes current assets, property, plant and equipment, intangibles, and other long-term assets. We construct the Debt Expansion Restraint signal by calculating the year-over-year change in long-term debt issuance, scaled by lagged total assets: $(DLTIS(t) - DLTIS(t-1)) / AT(t-1)$. This measure captures the acceleration or deceleration in a firm’s debt financing activities relative to its asset base. A positive value indicates that a firm is increasing its pace of debt issuance compared to the prior year, while a negative value suggests restraint in debt expansion. By scaling the change in debt issuance by lagged assets, we control for firm size and make the measure comparable across firms of different scales. We compute this signal using fiscal year-end values and require non-missing observations for both DLTIS and AT in consecutive years. Firms with zero or negative total assets are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the DER signal. Panel A plots the time-series of the mean, median, and interquartile range for DER. On average, the cross-sectional mean (median) DER is -0.03 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input DER data. The signal’s interquartile range spans -0.06 to 0.05. Panel B of Figure 1 plots the time-series of the coverage of the DER signal for the CRSP universe. On average, the DER signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does DER predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on DER using NYSE breaks. The first two lines of Panel A report

monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high DER portfolio and sells the low DER portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short DER strategy earns an average return of 0.26% per month with a t-statistic of 3.45. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.25% to 0.31% per month and have t-statistics exceeding 3.22 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.24, with a t-statistic of 4.66 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 535 stocks and an average market capitalization of at least \$1,410 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns

to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 21 bps/month with a t-statistics of 4.67. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for nineteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -5-20bps/month. The lowest return, (-5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -0.82. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the DER trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in seventeen cases.

Table 3 provides direct tests for the role size plays in the DER strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and DER, as well as average returns and alphas for long/short trading DER strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the DER strategy achieves an average return of 34 bps/month

with a t-statistic of 3.65. Among these large cap stocks, the alphas for the DER strategy relative to the five most common factor models range from 25 to 36 bps/month with t-statistics between 2.64 and 3.90.

5 How does DER perform relative to the zoo?

Figure 2 puts the performance of DER in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the DER strategy falls in the distribution. The DER strategy’s gross (net) Sharpe ratio of 0.48 (0.37) is greater than 87% (94%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the DER strategy (red line).² Ignoring trading costs, a \$1 invested in the DER strategy would have yielded \$3.21 which ranks the DER strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the DER strategy would have yielded \$1.92 which ranks the DER strategy in the top 7% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the DER relative to those. Panel A shows that

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

the DER strategy gross alphas fall between the 58 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The DER strategy has a positive net generalized alpha for five out of the five factor models. In these cases DER ranks between the 76 and 84 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does DER add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of DER with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price DER or at least to weaken the power DER has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DER} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DER trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on DER and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the DER signal in these Fama-MacBeth regressions exceed 0.31, with the minimum t-statistic occurring when controlling for change in net operating assets. Controlling for all six closely related anomalies, the t-statistic on DER is -1.22.

Similarly, Table 5 reports results from spanning tests that regress returns to the DER strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the DER strategy earns alphas that range from 22-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.88, which is achieved when controlling for change in net operating assets. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the

DER trading strategy achieves an alpha of 22bps/month with a t-statistic of 3.01.

7 Does DER add relative to the whole zoo?

Finally, we can ask how much adding DER to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the DER signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes DER grows to \$1243.77.

8 Conclusion

This study provides robust evidence that Debt Expansion Restraint (DER) represents a significant and economically meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that DER maintains substantial predictive power even after accounting for transaction costs and controlling for established risk factors and related anomalies.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which DER is available.

The key findings reveal that a value-weighted long/short strategy based on DER generates impressive risk-adjusted returns, with an annualized net Sharpe ratio of 0.37 and statistically significant abnormal returns relative to the Fama-French five-factor model plus momentum. Particularly noteworthy is the signal’s resilience when tested against a comprehensive set of related factors from the factor zoo, including net debt financing, changes in financial liabilities, and various measures of asset growth. The persistence of a 22 basis points monthly alpha (t -statistic = 3.01) after controlling for these closely related strategies underscores that DER captures unique information about future stock returns not subsumed by existing factors.

From a practical perspective, these results suggest that investors can potentially enhance portfolio performance by incorporating DER into their investment strategies. The signal’s economic significance remains substantial even after accounting for implementation costs, indicating its viability for real-world applications. The findings also contribute to the broader literature on the relationship between corporate financing decisions and stock returns, supporting the view that firms exhibiting restraint in debt expansion tend to outperform those with aggressive debt growth.

However, several limitations warrant consideration. First, while our analysis follows state-of-the-art protocols for factor validation, the true out-of-sample performance of DER remains to be tested as markets evolve and adapt. Second, the economic mechanisms underlying the DER anomaly require further theoretical and empirical investigation to fully understand why debt expansion restraint predicts returns. Third, the analysis is limited to U.S. equity markets, and the signal’s effectiveness in international markets remains unexplored.

Future research should address these limitations by examining DER’s performance across different market regimes, geographical regions, and asset classes. Additionally, investigating the interaction between DER and macroeconomic conditions, particularly interest rate environments and credit cycles, could provide valuable in-

sights into when the signal is most effective. Finally, exploring the behavioral and institutional factors that might explain why markets appear to systematically misprice firms based on their debt expansion policies would enhance our understanding of this anomaly and its persistence in increasingly efficient markets.

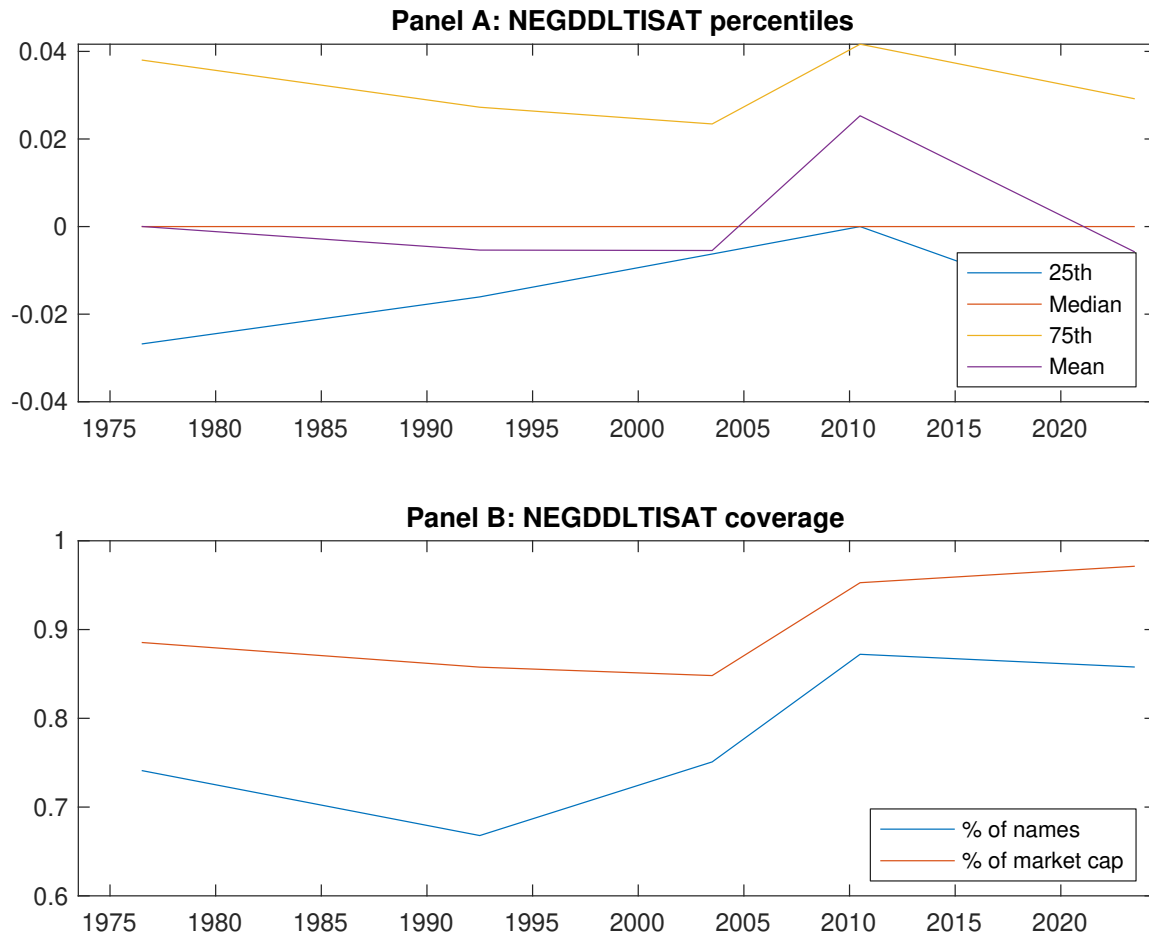


Figure 1: Times series of DER percentiles and coverage. This figure plots descriptive statistics for DER. Panel A shows cross-sectional percentiles of DER over the sample. Panel B plots the monthly coverage of DER relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on DER. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on DER-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.59 [2.81]	0.64 [3.53]	0.62 [3.13]	0.72 [4.04]	0.85 [4.26]	0.26 [3.45]
α_{CAPM}	-0.12 [-2.23]	0.03 [0.56]	-0.05 [-0.89]	0.11 [2.49]	0.18 [3.11]	0.30 [4.02]
α_{FF3}	-0.12 [-2.22]	-0.02 [-0.40]	-0.00 [-0.02]	0.09 [2.02]	0.19 [3.49]	0.31 [4.10]
α_{FF4}	-0.10 [-1.92]	0.01 [0.12]	0.04 [0.82]	0.06 [1.31]	0.17 [3.05]	0.27 [3.58]
α_{FF5}	-0.12 [-2.25]	-0.06 [-1.32]	0.05 [0.92]	0.01 [0.15]	0.15 [2.65]	0.27 [3.53]
α_{FF6}	-0.11 [-2.05]	-0.04 [-0.87]	0.08 [1.51]	-0.01 [-0.26]	0.13 [2.41]	0.25 [3.22]
Panel B: Fama and French (2018) 6-factor model loadings for DER-sorted portfolios						
β_{MKT}	1.06 [83.62]	0.98 [96.26]	0.99 [77.85]	0.98 [97.21]	1.02 [79.29]	-0.04 [-2.21]
β_{SMB}	0.16 [8.39]	-0.14 [-9.08]	-0.02 [-0.87]	-0.04 [-2.72]	0.14 [7.51]	-0.01 [-0.55]
β_{HML}	-0.03 [-1.21]	0.16 [8.06]	-0.10 [-4.23]	0.02 [1.22]	-0.14 [-5.53]	-0.11 [-3.16]
β_{RMW}	0.08 [3.16]	0.12 [5.75]	-0.05 [-2.01]	0.11 [5.53]	0.02 [0.82]	-0.06 [-1.66]
β_{CMA}	-0.09 [-2.35]	0.00 [0.02]	-0.11 [-2.90]	0.16 [5.38]	0.15 [4.09]	0.24 [4.66]
β_{UMD}	-0.02 [-1.24]	-0.03 [-3.09]	-0.05 [-4.05]	0.03 [2.78]	0.02 [1.46]	0.03 [1.95]
Panel C: Average number of firms (n) and market capitalization (me)						
n	671	535	1060	590	641	
me (\$10 ⁶)	1410	3195	2218	3114	1416	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the DER strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.26 [3.45]	0.30 [4.02]	0.31 [4.10]	0.27 [3.58]	0.27 [3.53]	0.25 [3.22]
Quintile	NYSE	EW	0.21 [4.67]	0.24 [5.27]	0.22 [4.91]	0.20 [4.46]	0.21 [4.59]	0.20 [4.32]
Quintile	Name	VW	0.25 [3.42]	0.29 [4.11]	0.30 [4.23]	0.26 [3.59]	0.25 [3.42]	0.22 [3.04]
Quintile	Cap	VW	0.21 [3.29]	0.25 [3.82]	0.25 [3.78]	0.20 [3.10]	0.17 [2.70]	0.15 [2.29]
Decile	NYSE	VW	0.23 [2.42]	0.31 [3.30]	0.28 [2.94]	0.24 [2.49]	0.12 [1.33]	0.11 [1.19]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.20 [2.64]	0.23 [3.13]	0.24 [3.17]	0.22 [2.91]	0.21 [2.76]	0.19 [2.54]
Quintile	NYSE	EW	-0.05 [-0.82]					
Quintile	Name	VW	0.19 [2.58]	0.24 [3.31]	0.24 [3.38]	0.22 [3.06]	0.20 [2.80]	0.18 [2.55]
Quintile	Cap	VW	0.16 [2.48]	0.20 [3.06]	0.19 [2.98]	0.17 [2.63]	0.13 [2.09]	0.12 [1.81]
Decile	NYSE	VW	0.16 [1.66]	0.24 [2.50]	0.21 [2.19]	0.19 [1.97]	0.08 [0.90]	0.07 [0.75]

Table 3: Conditional sort on size and DER

This table presents results for conditional double sorts on size and DER. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on DER. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high DER and short stocks with low DER. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
DER Quintiles						DER Strategies						
Size quintiles		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
	(1)	0.65 [2.36]	0.86 [3.20]	0.94 [3.41]	0.87 [3.18]	0.83 [2.91]	0.17 [1.86]	0.20 [2.16]	0.19 [1.99]	0.14 [1.44]	0.15 [1.62]	0.12 [1.28]
	(2)	0.72 [2.69]	0.90 [3.65]	0.82 [3.22]	0.88 [3.58]	0.87 [3.42]	0.14 [1.78]	0.18 [2.26]	0.15 [1.96]	0.16 [1.98]	0.14 [1.73]	0.14 [1.78]
	(3)	0.78 [3.06]	0.79 [3.62]	0.81 [3.35]	0.85 [3.79]	0.90 [3.83]	0.12 [1.52]	0.19 [2.41]	0.18 [2.28]	0.14 [1.75]	0.16 [1.98]	0.13 [1.62]
	(4)	0.73 [3.18]	0.78 [3.75]	0.84 [3.84]	0.74 [3.55]	0.91 [4.15]	0.17 [2.31]	0.21 [2.75]	0.20 [2.68]	0.18 [2.28]	0.19 [2.39]	0.17 [2.12]
	(5)	0.50 [2.56]	0.60 [3.36]	0.58 [2.94]	0.61 [3.35]	0.83 [4.42]	0.34 [3.65]	0.36 [3.88]	0.36 [3.90]	0.29 [3.15]	0.29 [3.13]	0.25 [2.64]
	Panel B: Portfolio average number of firms and market capitalization											
Size quintiles	DER Quintiles					DER Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)	
	(1)	392	393	393	393	389	37	34	32	33	35	
	(2)	107	107	107	108	107	61	61	60	61	61	
	(3)	76	77	76	76	77	107	110	105	107	108	
	(4)	64	65	64	65	64	229	240	229	235	230	
(5)	59	59	59	59	59	1453	2220	1800	2187	1517		

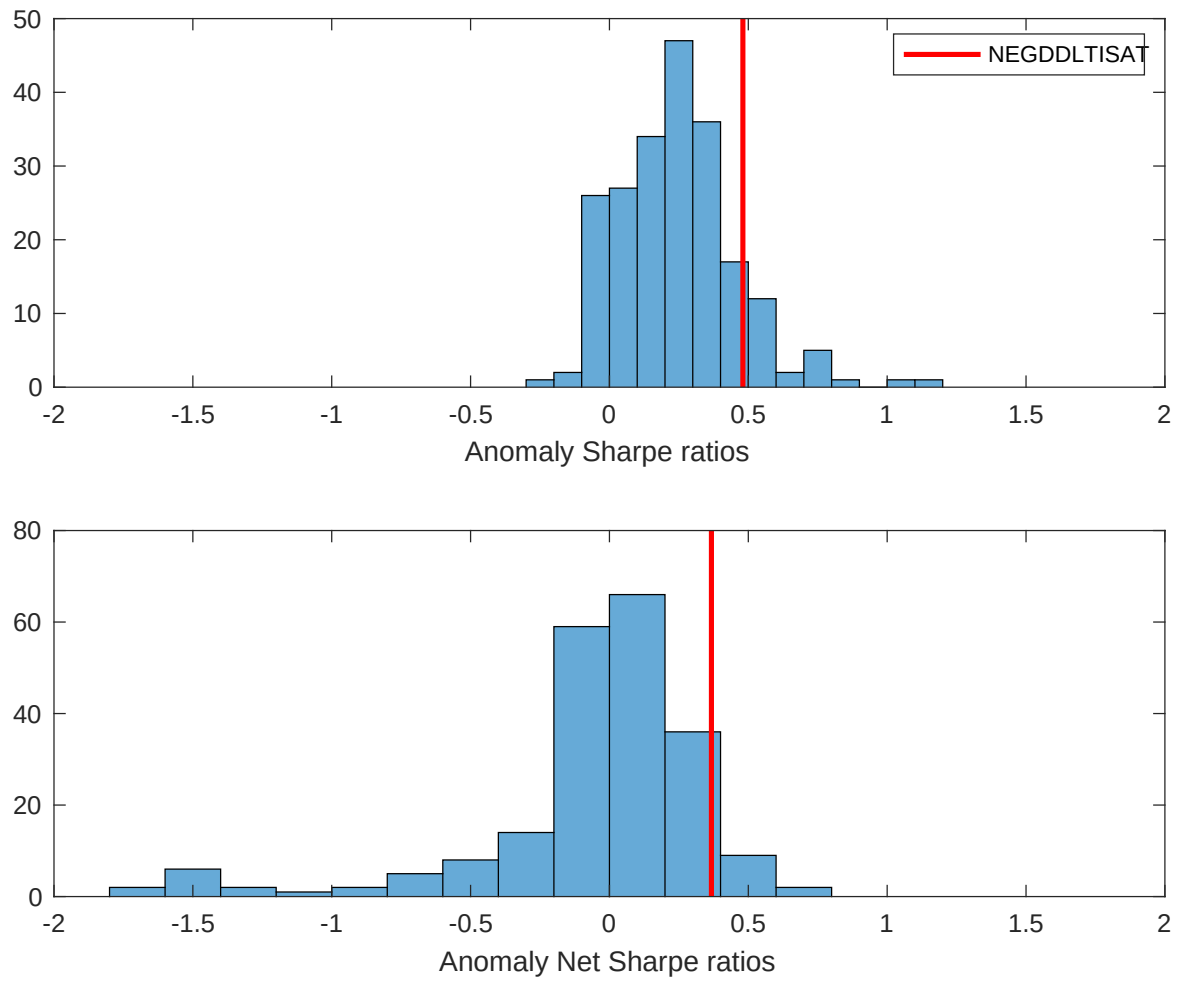


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the DER with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

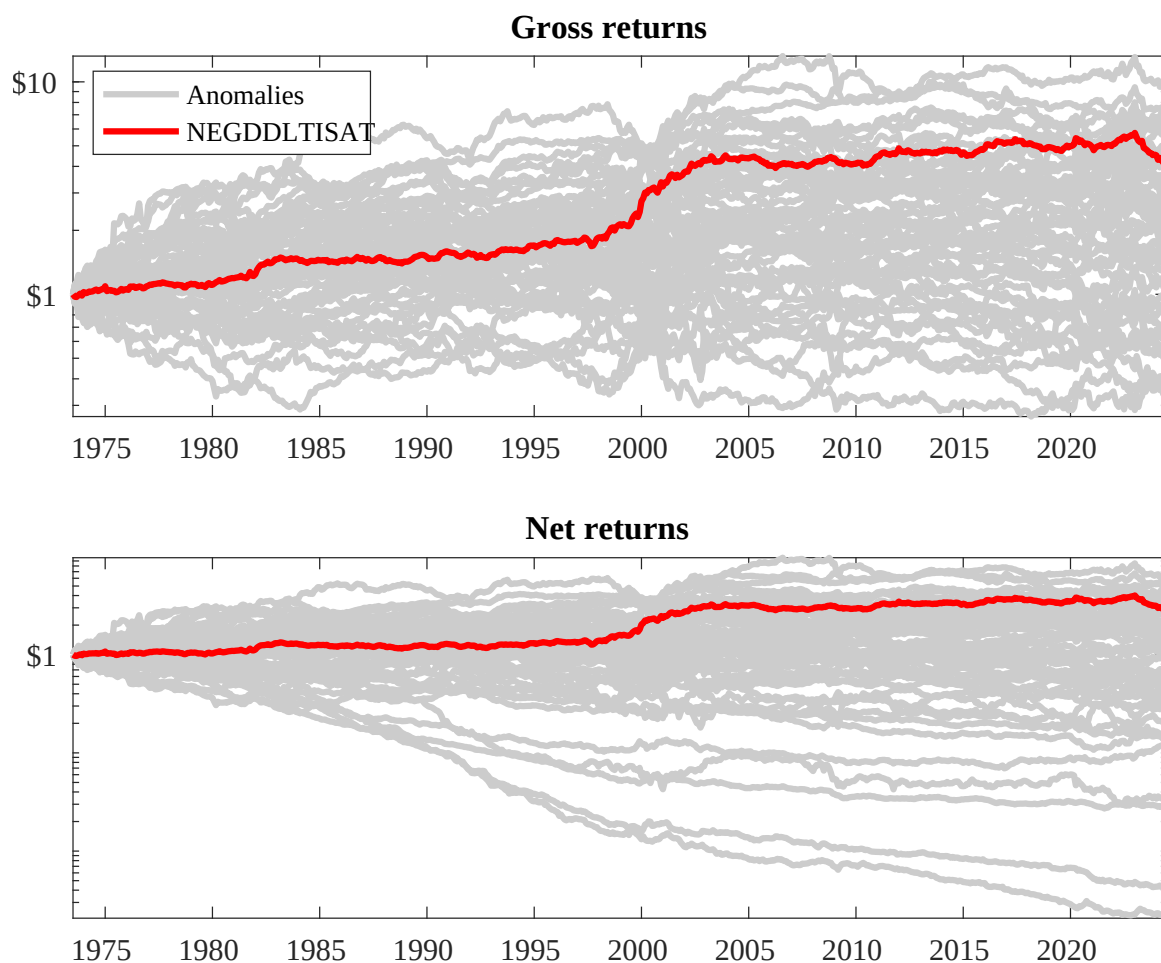
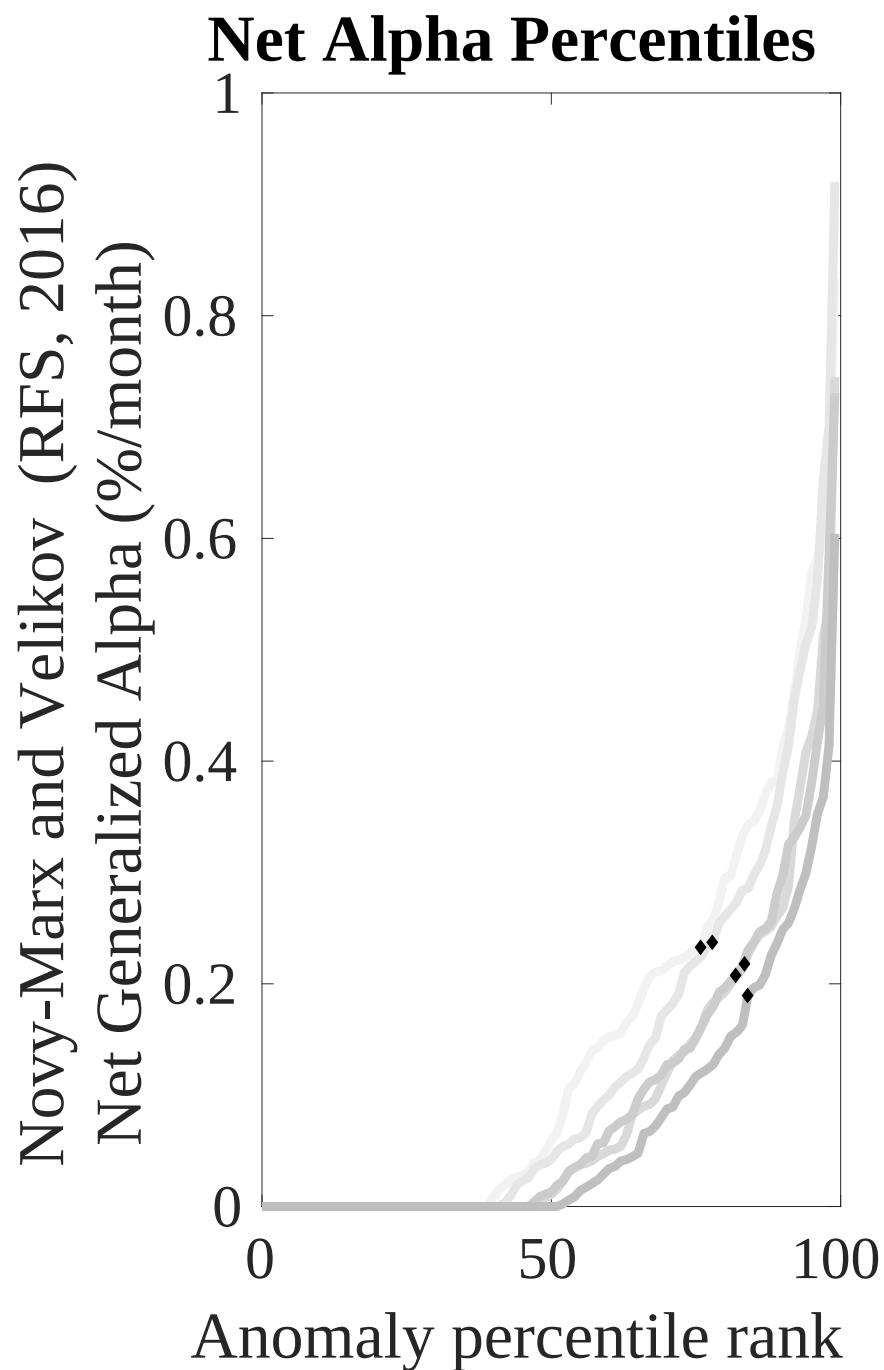
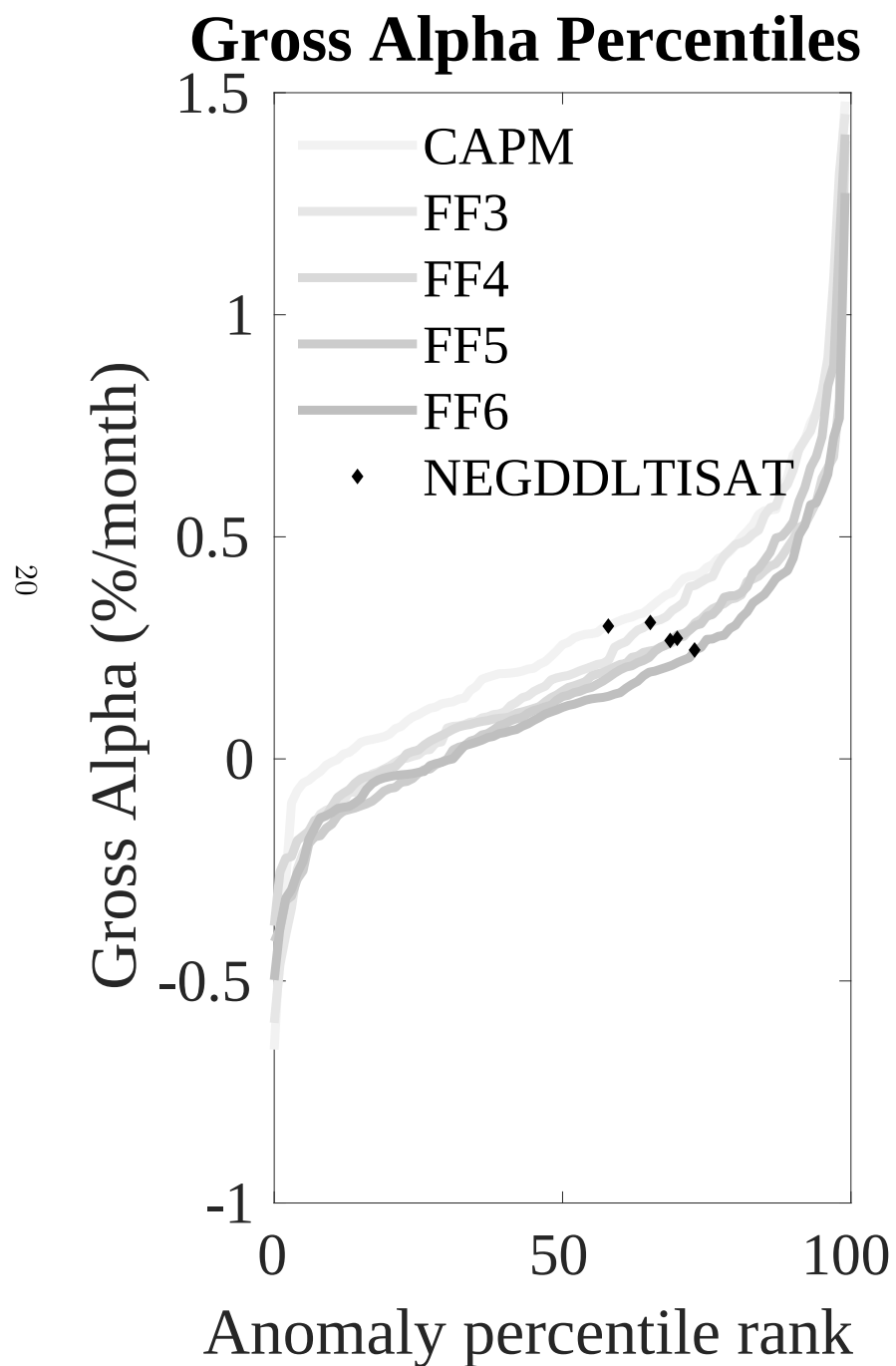


Figure 3: Dollar invested.

This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the DER trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.



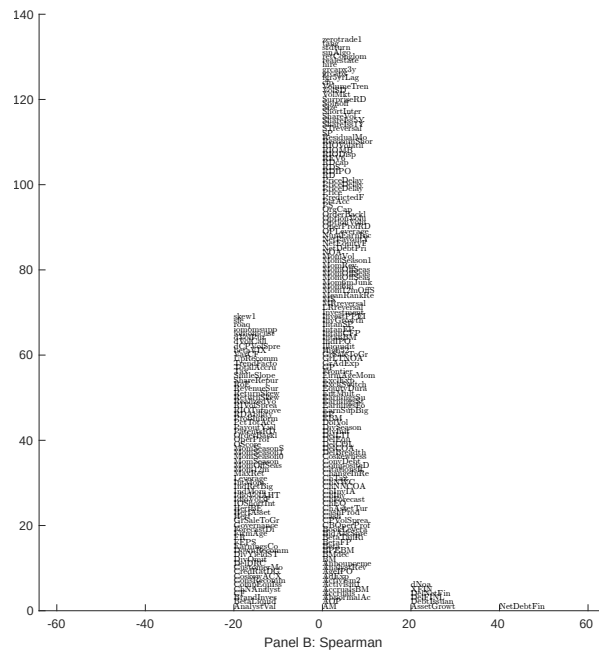
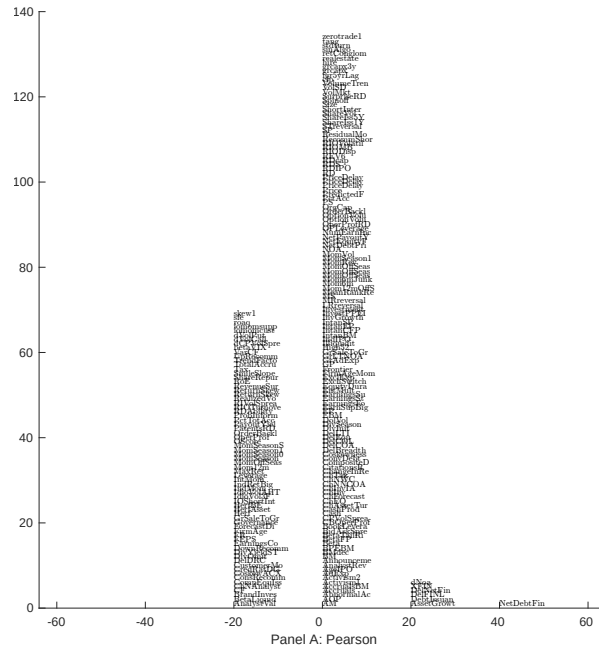


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with DER. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

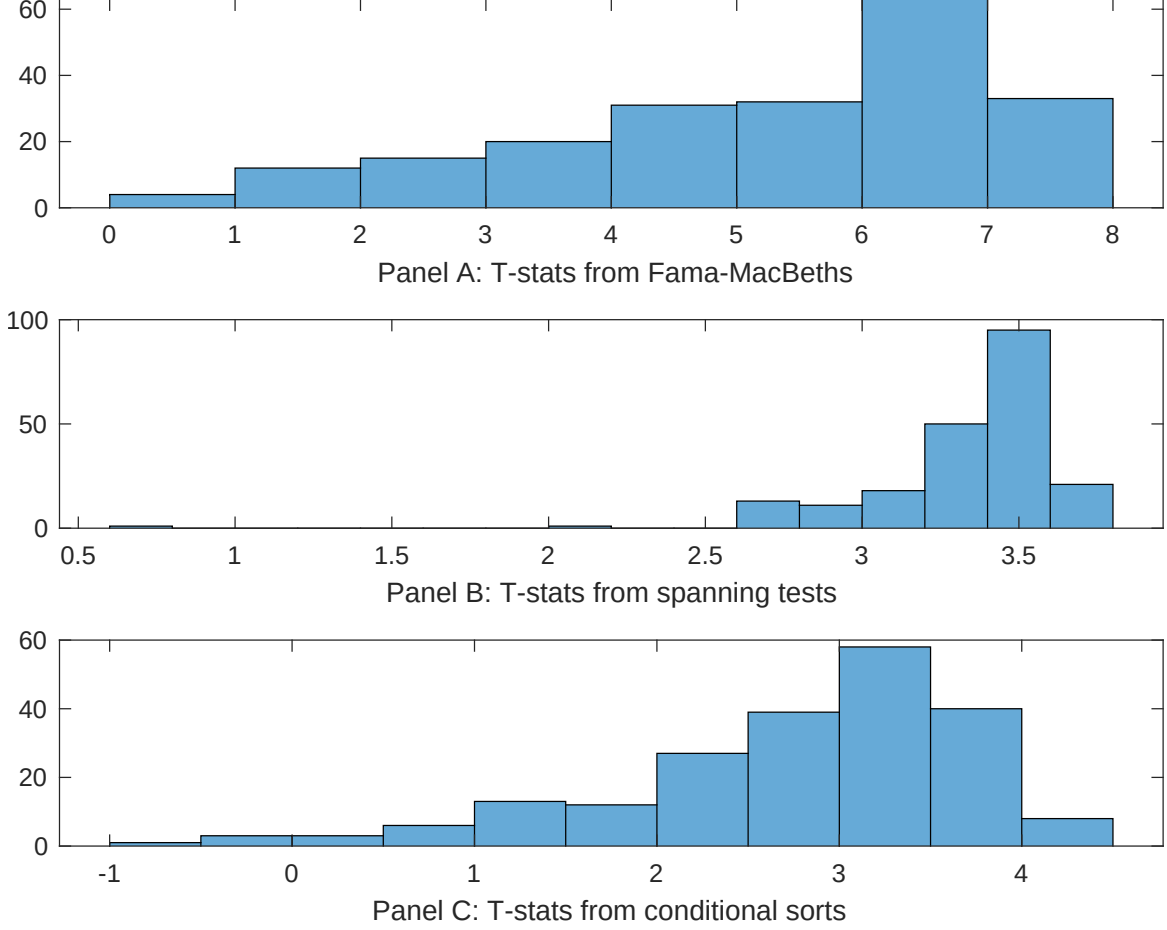


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of DER conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{DER} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{DER,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on DER. Stocks are finally grouped into five DER portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted DER trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on DER. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{DER}DER_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.33]	0.13 [5.36]	0.14 [5.32]	0.14 [5.69]	0.14 [5.73]	0.14 [5.65]	0.15 [5.75]
DER	0.12 [1.21]	0.10 [1.11]	0.51 [5.11]	0.20 [1.88]	0.21 [2.33]	0.29 [0.31]	-0.14 [-1.22]
Anomaly 1	0.20 [8.54]						0.11 [1.59]
Anomaly 2		0.18 [9.14]					-0.10 [-2.20]
Anomaly 3			0.40 [6.87]				0.21 [0.35]
Anomaly 4				0.19 [6.09]			0.11 [2.16]
Anomaly 5					0.17 [7.67]		0.69 [2.29]
Anomaly 6						0.14 [9.71]	0.96 [4.63]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	1	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the DER trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{DER} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Net debt financing, Change in financial liabilities, Inventory Growth, Net external financing, change in ppe and inv/assets, change in net operating assets. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.22 [2.94]	0.22 [2.88]	0.24 [3.20]	0.22 [2.89]	0.23 [3.12]	0.23 [2.99]	0.22 [3.01]
Anomaly 1	19.46 [4.77]						10.38 [1.87]
Anomaly 2		16.73 [3.94]					10.42 [1.79]
Anomaly 3			10.13 [3.46]				9.19 [2.99]
Anomaly 4				13.69 [3.64]			9.30 [2.36]
Anomaly 5					9.28 [2.72]		4.22 [1.12]
Anomaly 6						4.34 [1.00]	-8.15 [-1.68]
mkt	-3.52 [-2.05]	-3.17 [-1.84]	-3.74 [-2.16]	-1.68 [-0.93]	-3.68 [-2.12]	-3.54 [-2.03]	-2.26 [-1.27]
smb	-2.96 [-1.13]	-2.96 [-1.12]	-0.13 [-0.05]	3.16 [1.10]	-1.21 [-0.46]	-1.16 [-0.44]	0.63 [0.22]
hml	-9.73 [-2.94]	-9.41 [-2.82]	-10.85 [-3.25]	-9.00 [-2.68]	-11.52 [-3.42]	-10.88 [-3.21]	-8.46 [-2.52]
rmw	-7.67 [-2.26]	-7.04 [-2.07]	-4.06 [-1.19]	-13.71 [-3.37]	-5.36 [-1.57]	-5.46 [-1.60]	-11.89 [-2.93]
cma	19.28 [3.76]	18.73 [3.57]	15.33 [2.69]	15.40 [2.74]	16.98 [2.95]	21.12 [3.51]	6.63 [1.02]
umd	2.34 [1.36]	2.14 [1.22]	2.55 [1.46]	3.50 [2.04]	3.57 [2.07]	3.45 [1.98]	1.26 [0.72]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	10	9	9	9	8	7	13

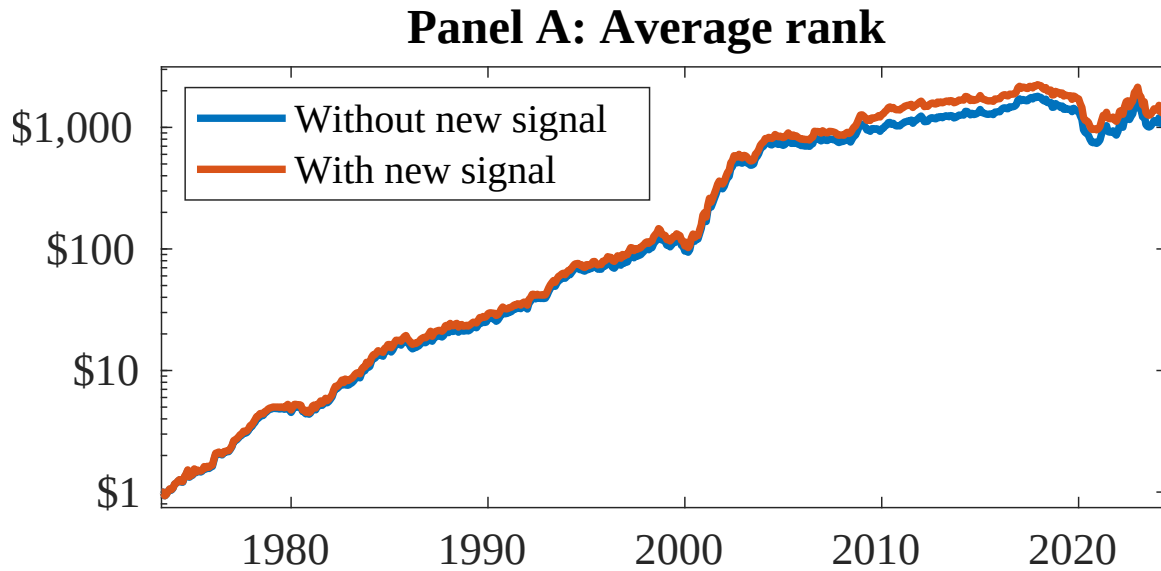


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as DER. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

- Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.
- Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.
- Breeden, D. T. (1979). An intertemporal asset pricing model with stochastic consumption and investment opportunities. *Journal of Financial Economics*, 7(3):265–296.
- Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.
- Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.
- Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.
- Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.
- Cochrane, J. H. (2005). *Asset Pricing*. Princeton University Press, revised edition.
- Cooper, M. J., Gulen, H., and Schill, M. J. (2008). Asset growth and the cross-section of stock returns. *Journal of Finance*, 63(4):1609–1651.
- Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance*, Forthcoming.

- Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.
- Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.
- Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.
- Hansen, L. P., Heaton, J. C., and Li, N. (2008). Consumption strikes back? measuring long-run risk. *Journal of Political Economy*, 116(2):260–302.
- Hou, K., Xue, C., and Zhang, L. (2015). Digesting anomalies: An investment approach. *Review of Financial Studies*, 28(3):650–705.
- Lettau, M. and Ludvigson, S. (2001). Consumption, aggregate wealth, and expected stock returns. *Journal of Finance*, 56(3):815–849.
- Livdan, D., Sapriz, H., and Zhang, L. (2009). Financially constrained stock returns. *Journal of Finance*, 64(4):1827–1862.
- Lyandres, E., Sun, L., and Zhang, L. (2007). The new issues puzzle: Testing the investment-based explanation. *Review of Financial Studies*, 21(6):2825–2855.
- Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.
- Novy-Marx, R. and Velikov, M. (2023). Assaying anomalies. *Working paper*.
- Wachter, J. A. (2013). Can time-varying risk of rare disasters explain aggregate stock market volatility? *Journal of Finance*, 68(3):987–1035.
- Whited, T. M. and Wu, G. (2006). Financial constraints risk. *Review of Financial Studies*, 19(2):531–559.